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RESEARCH ARTICLE

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# Evaluating strategies to increase PET bottle recycling in the United States

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## Abstract

In the United States, polyethylene terephthalate (PET) bottle collection rates have not increased in a decade. Recycling rates remain abysmal while industry commitments and policy targets escalate the demand for recycled plastics. We investigate the PET bottle recycling system, where collection is a critical bottleneck and recycled PET supply is not meeting the expected demand. We characterize demand for recycled PET (R-PET), analyze scenarios of expanding deposit return systems (DRS), and quantify cost barriers to improving PET bottle recycling. We find that a nation-wide DRS can increase PET bottle recycling rates from 24% to 82%, supplying approximately 2700 kt of recycled PET annually. With stability in demand, we estimate that this PET bottle recycling system can achieve 65% bottle-to-bottle circularity, at a net cost of 360 USD/tonne of PET recycled. We also discuss environmental impacts, stakeholder implications, producer responsibility, and complimentary policies toward an efficient and effective recycling system.

**KEYWORDS**
industrial ecology, market based, plastic waste, plastics recycling, polyethylene terephthalate (PET), recycled content

## 1 | INTRODUCTION

Staggering statistics (Borrelle et al., 2020; Geyer et al., 2017; Zheng & Suh, 2019) about plastic waste have driven multi-stakeholder engagement from entrepreneurs searching for innovative technological solutions (Ellis et al., 2021; Jehanno et al., 2022), packaging companies pledging circular measures (Ellen MacArthur Foundation, 2021), to nations setting ambitious recycling targets (OECD, 2022c). Despite these efforts, progress toward recycling outcomes has been slow. Two interrelated challenges impede plastics recycling: limited quantity of collected plastic waste and lower quality of the recycled polymer (Klotz et al., 2023). Put in economic terms, collection quantity constrains recycled supply, and without adequate quality, recycled demand suffers (Milios et al., 2018). For post-consumer plastics, the quantity of collection depends on the ease of consumer participation in recycling (Walzberg et al., 2023), while quality concerns in the recycled polymer can be traced to contamination and polymer degradation (Demets et al., 2021).

We focus our study on polyethylene terephthalate (PET) bottles, where commercial-scale recycling has been established for more than three decades (Stein, 1998; Welle, 2011). Several technological factors allay quality concerns in the recycling of PET bottles: bottles have a consistent, identifiable product format that can be accurately sorted; non-PET components of bottles (such as caps or labels) have easy separation processes

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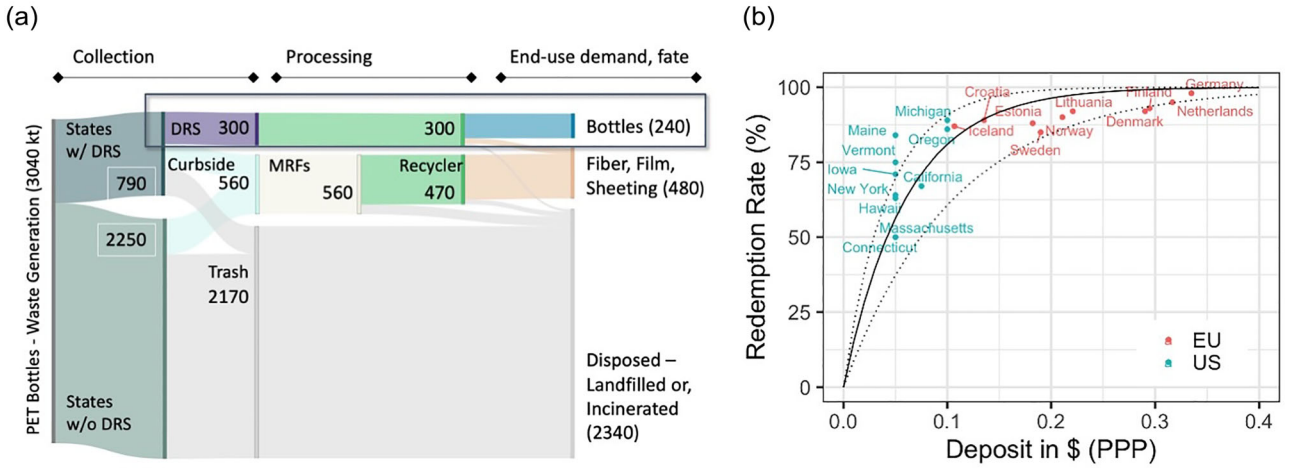



**FIGURE 1** Deposit return systems and polyethylene terephthalate (PET) recycling. (a) 2018 PET bottle material flow at end-of-life. DRS, deposit return system; MRF, materials recovery facility. Box highlights that of the 300 kt of PET currently collected via DRS in 2018, 240 kt is sold for use in bottles, and 55 kt is sold for non-circular applications. (b) Deposit return system redemption rate as a function of deposit fees for 10 states in the United States (blue) and 10 countries in the EU (red). The redemption rate denotes the fraction of eligible PET bottles returned by consumers to acceptable redemption locations. Data from 2019–2020 (Reloop Platform, 2020). PPP, purchasing power parity. Underlying data for Figures 1a and 1b can be found in Supporting Information S2, Tables A3 and A4, respectively.

(Mantia, 2002); and PET degradation can be partially reversed by solid-state polycondensation (Welle, 2011). In fact, recent news reports suggest that as much as 100% recycled content can be incorporated into many PET bottles made by major beverage brands (Doering, 2021). However, despite overcoming the many quality barriers, PET bottle recycling rates in the United States have been stagnant or slightly declining. NAPCOR, a trade association for PET containers, reports that while 31% of PET bottles placed on the market were collected for recycling in 2011, only 28.6% were collected in 2021 (NAPCOR, 2021). Moreover, US PET bottle collection rates are far lower than many countries in Europe (>90% in Denmark, Germany, Belgium, Finland, and Lithuania) and Asia (87% in Japan and 80% in India) (OECD, 2022a). Collection presents a critical bottleneck to improving the US PET bottle recycling system (US EPA, 2021).

Two key collection modes underpin PET bottle recycling in the United States. Curbside collection, available to 73% of residents across all states (The Recycling Partnership, 2020), represents locally managed collection programs where PET bottles are collected amongst other mixed recyclables and sorted at material recovery facilities (MRFs). In addition to curbside collection, 10 states have deposit return systems (DRS) where eligible PET bottles can be returned to acceptable redemption locations such as participating retail centers or dedicated deposit centers. A deposit fee is charged upfront during the sale of the beverage and returned when bottles are returned. In a deposit return scheme, PET bottles are source separated from other packaging products like aluminum cans and glass bottles (National Council of State Legislatures, 2022). Material flow estimates (Figure 1a) suggest that in 2018, the PET bottle recycling rate, after accounting for losses along the recycling value chain, is 24%, and the bottle-to-bottle circularity rate is 11% (the collection rate is 28%). With 2340 kt (or 76%) of PET bottle waste generated disposed in landfills or incinerated in waste-to-energy facilities, there is a massive opportunity for efficiency and improvement in the US PET bottle recycling system.

Evidence from DRS (Zhou et al., 2020) in many countries around the world as well as in several states within the United States makes it an effective candidate for increasing collection. As shown in Figure 1a, the percentage of PET waste generated that is sent to trash is 34% for the 10 states with DRS compared to 83% for states without. Moreover, as shown in Figure 1b, higher deposit fees correlate with higher redemption rates in DRS states (and several EU nations), suggesting a means to improve PET bottle recycling rates. DRS collection also provides a cleaner waste PET supply stream (Eriksen et al., 2019) that can be readily used as feedstock for bottle-to-bottle recycling with fewer separation processes and more stringently controlled contamination levels than curbside collection (Eriksen, 2019). On the other hand, curbside collected PET bottles are mostly used in non-bottle applications due to contamination and incur greater sorting losses.

Broader literature on environmental analysis of PET waste (Brogaard et al., 2014; Shen et al., 2011; Smith et al., 2022) acknowledges the benefits of recycling and recommends increased recovery (Lau et al., 2020; Meys et al., 2021; Zheng & Suh, 2019). But the means and costs of achieving higher recycling rates are seldom discussed. Policies such as recycled content mandates (De Smet et al., 2019) aim to boost circular recycling practices by setting targets for beverage producers and creating demand for high-quality R-PET (Carriere & Beavers, 2020). In the United States, the proposed "Break Free from Plastics Pollution Act of 2021" outlines recycled content targets of 25% by 2025, 50% by 2030, 70% by 2035, and 80% by 2040 (Break Free From Plastic Pollution Act, n.d.). While the future of this bill or allocated enforcement capacity is not certain, California has already instituted the same targets (California Beverage Container Recycling and Litter Reduction Act, n.d.) for 2025 and 2030 and several major beverage producers have also made recycled content commitments (Ellen MacArthur Foundation, 2021) that are aligned with the 2030 trajectory. In fact, for the near term to 2025, not enough PET bottles are being collected to meet this policy-driven demand for R-PET (Kahlert & Bening, 2022;

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Schneider, 2021). We assess the costs of increasing supply through the expansion of DRS and investigate the distribution of these costs among various stakeholders.

Previous reports (Edwards et al., 2019; Edwards & Carhart, 2020) have investigated DRS economics on the municipal or state level, and their results suggest net savings for the local government. But the narrow geographical scope of such studies limits incentivizing large-scale action in a system of PET reclaimers and end-use markets that are connected at a national level (Smith et al., 2022). However, detractors of DRS cite concerns (Waste Dive, 2022) about the cost of the policy, consumer inconvenience, and adverse economic effects on materials recovery facilities (MRF) that process single-stream recyclables. Such limitations and concerns point to a larger gap in the plastics recycling literature filled by this contribution—currently, there is no framework that evaluates how plastics policy and recycling markets interact. Our analysis shows that by framing PET bottle recycling in supply and demand terms, we can better understand recycled plastics markets whose functioning is dependent on policy drivers.

Academic, policy, and practitioner literature alike agree that understanding and stimulating the market for recycled plastics is necessary (OECD, 2018; Olatayo et al., 2023; US EPA, 2021) to improve recycling outcomes. But market-based studies are rare. Only one study explicitly uses linear demand and supply curves to understand the impact of China's waste import ban on plastics recycling in Japan—but the authors only infer qualitative trends (Kumamaru & Takeuchi, 2021). In most economic studies of plastic waste recovery, the waste sorting (Cimpan et al., 2016) and recycling options (Larrain et al., 2021; Singh et al., 2021; Volk et al., 2021) are investigated independent of common collection models (Valenzuela et al., 2021) and remain divorced from downstream demand conditions (Kahlert & Bening, 2022). A detailed system-wide environmental and socioeconomic study (Bassi et al., 2022) of PET recycling in the European Union (EU) analyzes future scenarios and highlights the role of demand and secondary markets. However, the study only considers fixed relative market sizes and does not include the price effects of demand. Studies investigating the recycling of metals and paper demonstrate that understanding supply–demand interactions is critical to informing market-based policies (Mansikkasalo et al., 2014; Söderholm & Ekvall, 2020). Material flow analysis maps plastic waste generated, recycled, disposed, or leaked into the environment in various geographies (Heller et al., 2020; Kawecki et al., 2018; Lee et al., 2021; Nakatani et al., 2020) and comments on the potential for circular practices (Drewniok et al., 2023; Klotz et al., 2022), but lacks the necessary product specificity and economic considerations to enact such practices.

We adopt a supply–demand perspective to address these gaps and analyze strategies that can improve PET bottle recycling rates in the United States. We demonstrate how expanding DRS in the United States can meet recycling and circularity targets. We also estimate the costs of expansion under various demand scenarios and inform ongoing efforts to design and implement extended producer responsibility (EPR) for plastic packaging. On a methodological note, we illustrate how reframing plastics recycling barriers as recycled supply and demand considerations can quantify trade-offs across stakeholders for better decision-making.

## 2 | METHOD OVERVIEW

### 2.1 | Building supply curves for R-PET

In resource economics, material production costs and quantities are aggregated and visualized as supply cost curves. Such curves can then be used to approximate the cost of obtaining excess supply needed to meet demand. When waste is considered a resource for recycling, similar methods can be applied to understand policies and economics of waste recovery (Calcott & Walls, 2005; Porter, 2010). In this work, we model the US PET recycling system—including collection, sorting, and recycling, to build bottom-up supply cost curves for PET bottles. The supply curve orders all material suppliers in increasing order of aggregated cost, which includes waste collection and processing steps (Figure 2a). The cumulative quantity supplied forms the horizontal axis, while the marginal unit cost of each supplier forms the vertical axis. The width of each bar is the quantity produced by a supplier (e.g., a sorting facility) and the height of each bar is the cost of the supplier per tonne of final recycled output. In the United States, DRS legislation is a state-level policy action, while curbside collection parameters vary widely between counties and municipalities within a state. We include the necessary granularity to capture state-level and local-level details. We estimate deposit-based collection costs from state-level DRS handling fees (see S1.1.1). We regress a relationship between the cost to collect curbside mixed recyclables based on census variables and local policy (detailed in S1.2.1). We establish the costs of sorting, recycling, and transportation through process modeling, facility location, and distance optimization (see S1.2 and S1.3). An ordered list of these concatenated costs, disaggregated by R-PET supplier (e.g., municipality or reclaimer), creates a DRS and curbside supply cost curve (Figure S1) that informs supply economics. We use data from 2018 to 2020 to characterize the supply curve and summarize this data along with methodological details in the Supporting Information.

### 2.2 | Estimating demand curves for R-PET

R-PET demand curves abstract expected changes in recycled material consumption as a function of price. The price elasticity of R-PET demand indicates sensitivity to price, and prices reflect primary and secondary market conditions (Ryter et al., 2022). When R-PET price decreases, demand

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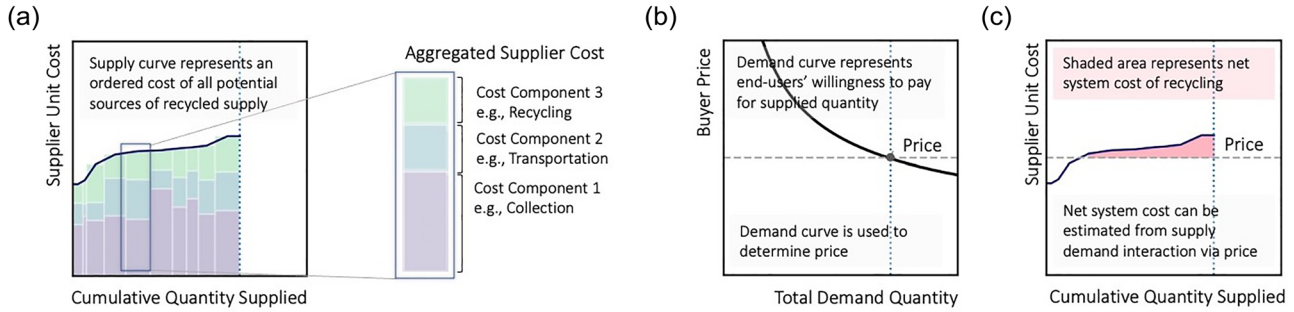



**FIGURE 2** Supply–demand interaction methodology. (a) Supply cost curve schematic with cost components. (b) The demand curve represents the relationship between R-PET price and the quantity of R-PET demanded by consumers at that price; the dotted vertical line is the total cumulative R-PET supply inferred from the supply curve. Price is estimated from the demand curve and is represented by the horizontal line. (c) Supply–demand interaction is used to evaluate net system costs. The net system cost is the area shown in pink and represents the difference between the unit cost and R-PET price for all loss-making suppliers.

increases as buyers are willing to purchase more recycled material. Therefore, price and quantity have an inverse relationship. However, demand for recycled PET is not only dependent on virgin PET markets but is increasingly driven by policy pressures as well (OECD, 2018).

To accommodate both market and policy factors, we model demand for R-PET as a function of virgin PET prices, and recycled content requirements that mandate minimum recycled content usage in PET bottles. Lower virgin PET price lowers R-PET demand and greater recycled content commitments increase demand. This functional relationship and related data are further elaborated in Supporting Information S1, Section S2. We also include quality considerations by delineating demand by grade: bottle end-users need high-quality (food-grade) R-PET and therefore limit their use to deposit-sourced recycled material while non-bottle end-users can use any R-PET source available. Correspondingly, two R-PET prices are estimated—a food-grade R-PET price and a low-grade R-PET price.

## 2.3 | Supply–demand interaction in future policy scenarios

Recycled material supply from post-consumer sources is largely price inelastic (Mansikkasalo et al., 2014); this is because municipal waste generation and recycling activities are pursued independent of recycling market forces. In the case of PET bottles, where post-consumer recycling is often a municipal service, consumers are unaffected by PET bale prices. Moreover, as a critical component of a city or state's waste management strategy, post-consumer material recovery is subsidized through public funds when unprofitable. Therefore, in our methodology, suppliers determine R-PET quantity (Figure 2a), and buyers determine the price at which the total supplied R-PET is purchased (Figures 2b,c). When more R-PET quantity is supplied than buyers are willing to purchase at a given price, R-PET prices fall until demand matches supply.

We analyze hypothetical supply increases via DRS expansion to meet the demand set by policy-based recycled content targets (25% recycled content by 2025, 50% by 2030, and 70% by 2035). The combination of demand- and supply-side policies form a "circularity roadmap" for increasing PET recycling. Figure 3a visualizes these scenarios. First, in 2025, all nine states with pre-existing DRS programs in the contiguous United States increase the value of the deposit fee to 10c and include all PET bottles under the program. In 2030, the remaining 39 states implement a 5c deposit fee on all bottles, and in 2035, all states have a 10c deposit fee on all PET bottles. This combines a "national beverage container program" with a refund value "not less than 10c," as proposed by the Break Free from Plastics Pollution Act (Break Free From Plastic Pollution Act, n.d.).

## 2.4 | Metrics used to evaluate future policy scenarios

We evaluate scenarios across four metrics: net system costs, R-PET cost gap, circularity rate, and recycling rate. Net system cost estimates the total subsidy needed for both the curbside and deposit systems to meet all the costs that exceed the revenue from the sale of recycled products. The average unit cost of producing R-PET is partially met by the R-PET price (paid by R-PET buyers). The difference between this average R-PET cost (per tonne) and the price is the R-PET cost gap. In the long run, if the R-PET cost gap widens and is not adequately subsidized, components of the wider recycling system can risk closure. Circularity rate is the proportion of generated bottle waste that is recycled into bottle end-use. The recycling rate is the proportion of generated bottle waste that is recycled for use across both bottle and non-bottle end uses. Taken together, these four metrics summarize our policy scenarios and explain the costs and effectiveness of implementing demand-side policy strategies in stimulating the R-PET market.

We distinguish between the recycling rate and collection rate in this study. Because not all collected plastic waste is recycled, the recycling rate is lower than the collection rate due to sorting and other losses in the reverse supply chain. We also distinguish between recycling rate and recycled

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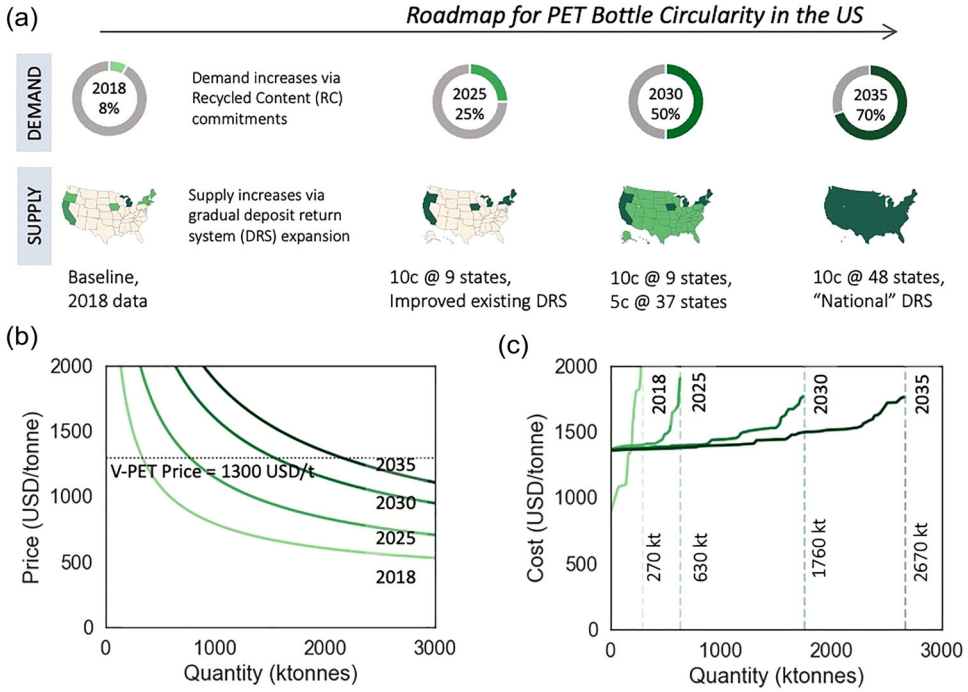



**FIGURE 3** Roadmap for polyethylene terephthalate (PET) bottle circularity in the United States. (a) Circularity roadmap includes recycled content targets and staged deposit return systems (DRS) expansion to 48 contiguous states in the United States. Detailed scenario supply and demand parameters are listed in Table S2 and S5, respectively, in SI. Recycled content targets are informed by proposed federal legislation in Break Free from Plastic Pollution Act (Break Free From Plastic Pollution Act, n.d.). (b) Demand evolution based on increasing recycled content targets. A long-term average of 1300 USD/tonne is used to construct bottle end-use demand for 2025, 2030, and 2035. (c) Deposit supply curves for DRS expansion scenarios. Underlying data for Figures 3b and 3c can be found in Supporting Information S2, Tables A5 and A6, respectively.

content: recycling rate is a supply-side metric describing the fraction of generated waste that is recycled. On the other hand, recycled content is a demand-side metric that accounts for the fraction of recycled material in a product such as PET bottles or clothing. Because recycled PET from bottles is used across multiple product categories, recycled content in one product category such as bottles is always lower than the total recycling rate.

## 3 | RESULTS

We visualize demand and supply cost curves for the baseline and future hypothetical scenarios in Figure 3b,c. We use system costs and recycling rates in 2018 for calibration and report projected results against the 2018 baseline. First, in Section 3.1, we estimate the recycling rates, circularity rates, R-PET price, and R-PET cost gap for the baseline and three DRS expansion scenarios in the PET circularity roadmap. The success of supply-side interventions such as DRS critically depends on sustained demand for R-PET, as demand determines the R-PET price, and R-PET price impacts net system costs. Because future demand is highly uncertain, we also explore the sensitivity of our results for the 2035 nation-wide DRS scenario to key demand factors. In Section 3.2, we vary virgin PET price and recycled content requirement for circular bottle end-use, and in Section 3.3, we study the effect of increased demand for non-circular end-uses such as recycled polyester. An important consideration in implementing DRS is the negative impact on MRFs that rely heavily on PET and aluminum bales for revenue and are vital components of the overall recycling system (Waste Dive, 2022). In Section 3.4, we estimate the impact of DRS implementation on MRF profitability. Finally, EPR for plastic packaging has increasingly become central to the discussion of recycling costs and economic viability in many states in the United States (Sustainable Packaging Coalition, 2022). In Section 3.5, we quantify the EPR fees that producers need to be charged in different scenarios such that the fees cover the costs of recycling PET bottles.

### 3.1 | PET circularity roadmap scenario cost analysis

In 2018, 720 kt of R-PET was placed on the market (recycling rate is 23.6%), and only a third (240 kt) was used for bottle end-use (Figure 4a). We find PET recycling unprofitable at a total net system cost of 210 million USD (Figure 4b). Handling fees for deposit centers as well as municipal collection

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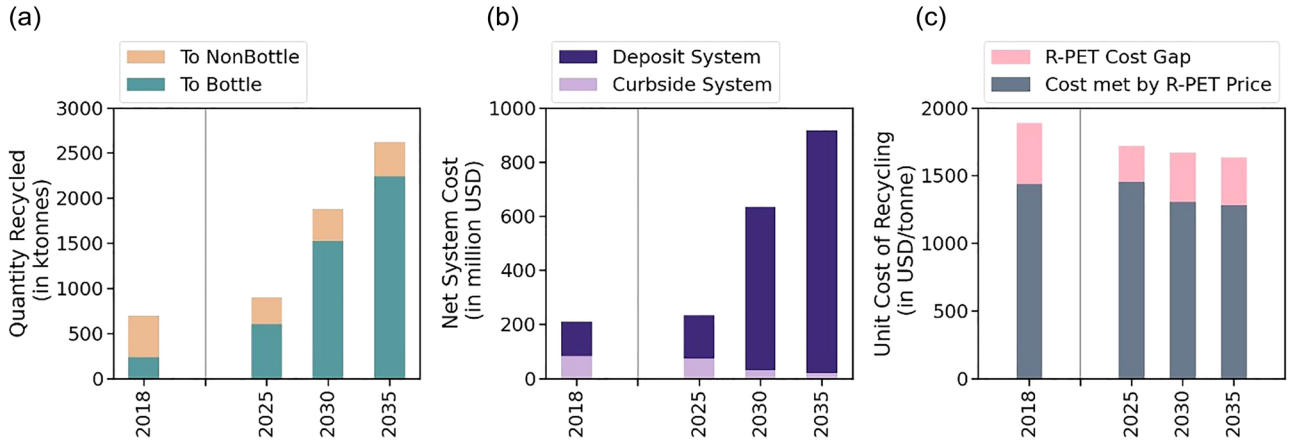



**FIGURE 4** Deposit return systems expansion quantities and cost. (a) Quantity recycled to bottle, and non-bottle end uses. (b) Net system cost for curbside and deposit channels in 2018, 2025, 2030, and 2035 scenarios. (c) Food-grade R-PET price and R-PET cost gap; R-PET cost gap is the difference between the average cost of supplying R-PET and R-PET price determined by demand. The virgin polyethylene terephthalate price is $1600/tonne in 2018 and USD 1300/tonne in future scenarios. The underlying data for this figure can be found in Supporting Information S2, Table A7.

and processing fees paid to MRFs are the implicit subsidies that balance this cost burden. Food-grade R-PET price of 1430 USD/tonne is lower than the V-PET price (1600 USD/tonne), and a R-PET cost gap of 450 USD/tonne is estimated for the deposit-collected R-PET (Figure 4c).

Implementation of state-wide DRS expansion and fee increase drives up R-PET supply and recycling rates (Figure 4a). The PET recycling rate increases from 23.6% in 2018 to 82% in 2035 (PET recycled quantity increases from 720 kt to almost 2650 kt) and the circularity rate improves more than 8× (8% to 68%). Demand from non-bottle end-use is kept constant and as a result, the fraction used for non-bottle end-use declines from 67% in 2018 to 14% in 2035 (Figure 4a). Due to the higher quantity of recycling in the national DRS scenario, annual net system cost increases from 210 million USD in 2018 to 920 million USD in 2035 (Figure 4b). However, we find that the cost per tonne of PET recycled is lowered (Figure 4c), due to economies of scale. The recycling system still requires subsidies, but the R-PET cost gap is lower (down from 450 USD/tonne in 2018 to 360 USD/tonne in 2030, 2035).

## 3.2 | Sensitivity to PET bottle demand factors

The trade-off between the R-PET price paid by buyers and the R-PET cost gap, usually covered by the government, is modulated by R-PET demand (Figure 5a). R-PET demand may fall short of the targets outlined in the circularity roadmap, due to (a) the availability of low-cost and consistently high-quality virgin PET and (b) changes in recycled content targets set by policy. Virgin PET (V-PET) prices are volatile—between 2018 and 2021, prices have varied between 1000 and 1800 USD/tonne (ICIS Chemical Business, 2019). As the virgin sector loses market share to recycled products, V-PET prices may decrease, eroding R-PET demand and increasing net system costs. Results suggest that net system costs can more than double from 920 million USD (Figure 5b, Case 1) to 1.94 billion USD (Figure 5b, Case 4) if V-PET prices near 1000 USD/tonne and the recycled content target drops to 50% in the 2035 nation-wide DRS scenario. In such cases, the R-PET price paid by buyers only covers 55% of the cost of collecting and processing R-PET (Figure 5c), and a cost gap of 770 USD/tonne is expected. Maintaining demand for R-PET is critical to cost-effectively improving recycling and circularity rates. Policies such as strong recycled content mandates and taxes on virgin material usage support R-PET demand when voluntary commitments fail, or virgin prices fall.

## 3.3 | Sensitivity to recycled polyester demand

Globally, polyester makes up over 50% of the fiber used in textiles (Textile Exchange, 2022) and is chemically similar to PET used in beverage bottles. Sustainability commitments made by clothing brands (Textile Exchange, 2021) can increase R-PET demand from bottles, as textile-to-textile recycling remains underdeveloped (Juanga-Labayen et al., 2022). We simulate a 100% increase (relative to 2018) in demand for R-PET fiber for the textile sector, competing with circular bottle-to-bottle recycling. We find that the resulting increase in R-PET price can cover the costs of the DRS expansion to a greater extent and bring down the total subsidy needed from 920 to 508 million USD (Figure S9). However, the circularity rate drops from 70% to 52% in the 2035 nation-wide DRS scenario. When multiple end markets exist, competing interests can distort the effect of circular policy tools. For instance, a recycled content mandate for bottles may lose its utility if non-bottle end uses capture most of the available supply. Bottle

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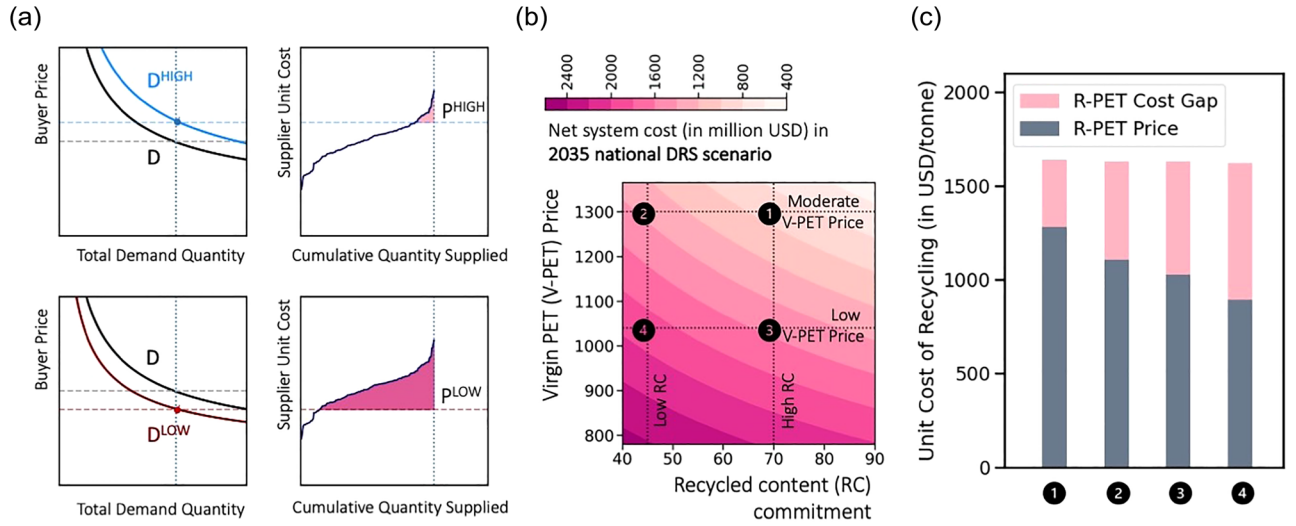



**FIGURE 5** Sensitivity to demand factors. (a) Schematic showing the effect of changing demand on R-PET price and net system cost of recycling. When R-PET demand is high ($D^{HIGH}$ compared to D), the R-PET price for a given quantity of supply increases. Higher R-PET price $P^{HIGH}$ covers more of the cost of supply and decreases the net system cost. (b) Total net system cost (in million USD) required to support demand for 2035 nation-wide deposit return systems expansion supply scenario under different virgin prices and recycled content (RC) commitments. (c) Unit cost of recycling (per tonne of R-PET produced) differentiated into R-PET price and R-PET cost gap for selected demand cases: Case 1 = 70% RC, 1300 USD/tonne V-PET price; Case 2 = 45% RC, 1300 USD/tonne V-PET price; Case 3 = 70% RC, 1040 USD/tonne V-PET price; and Case 4 = 45% RC, 1040 USD/tonne V-PET price. Underlying data for Figures 5b and c can be found in Supporting Information S2, Tables A8 and A9, respectively.

manufacturers concerned about rising competition for material can reduce their materials costs and increase recycled content rates by supporting EPR policies that allow them to retain ownership of post-consumer PET (OECD, 2022a).

## 3.4 | Impact on curbside collection and MRFs

When a state institutes a DRS, residents are more likely to take the PET bottle back to the retail store or a dedicated deposit center instead of placing it in the curbside recyclables collection bin. This effect is already seen in states that currently have DRS—only between 2% and 9% of beverage bottles are collected via curbside programs in the 10 states with DRS programs (Container Recycling Institute., 2022). As deposit systems expand, curbside PET bottle and aluminum can quantities destined for MRFs shrink. Since MRFs rely heavily on PET and aluminum bales for revenue, many have expressed concerns (Waste Dive, 2022) about DRS. If expansion of deposit programs leads to MRF shutdowns, DRS can impact the recycling of other materials in the curbside stream as well. To investigate this unintended consequence, we estimate the net unit processing costs (MRF costs − MRF revenue normalized by waste input mass) for all MRF-based recyclables (Figure S13). We find that, on average, net processing costs increase by 13% from 54 USD/tonne in 2018 to 61 USD/tonne in the 2035 scenario. Current processing fees that municipalities pay MRFs to cover the net unit processing cost range between 40 and 80 USD/tonne (The Recycling Partnership, 2020). Figure S13 also suggests that in 2035, smaller MRFs have larger increases in costs than larger MRFs. However, the total excess cost of keeping MRFs viable by subsidizing the additional cost increase is approximately 373 million USD. Identifying this potential externality and recognizing the cost burdens of various stakeholders (OECD, 2022a) in the interconnected recycling value chain when crafting legislation for DRS expansion is crucial.

## 3.5 | Extended producer responsibility fees

Currently, the net cost burden of recycling is borne by taxpayers in many states: municipalities subsidize collection and MRF processing while states often pay DRS handling fees to fund retailers and deposit centers. Packaging EPR laws, already set up in four states (Colorado, Maine, Oregon, and California) with varying degrees of stringency and scope (Sustainable Packaging Coalition, 2022), seek to transfer the externalized cost burden of waste management and recycling back to packaging producers. Implementation of EPR laws can enforce a financial obligation from producers in the form of EPR fees. At a minimum, the EPR fees must be designed to cover the R-PET cost gap we calculate in our analysis.

For the 2035 scenario, we estimate the value of EPR fees required to cover the cost gap (not including administrative charges) as a function of virgin price change and recycled content commitments (Figure 6). We find that EPR fees lie in the range of 300–700 USD/tonne for the four cases

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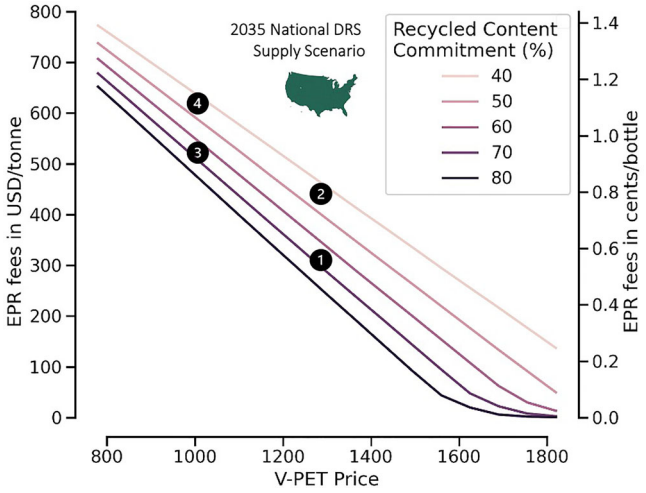



**FIGURE 6** Extended producer responsibility (EPR) fees. Estimated EPR fees (expressed both in USD/tonne and cents/bottle) to cover total net system costs for the 2035 nation-wide deposit return systems expansion scenario as a function of virgin polyethylene terephthalate (PET) price and recycled content commitments. Assumed average weight of a PET bottle = 31.7 g (see Section S3.3). The underlying data for this figure can be found in Supporting Information S2, Table A10.

highlighted in Figure 5b,c. This cost falls within the wide range of EPR fees implemented in Europe: 330 EUR/tonne for virgin PET bottles in 2021 by France's CITEO (CITEO, 2021); 104 EUR/tonne for transparent colorless PET bottles, and 596 EUR/tonne for transparent colored PET bottles in 2022 for Belgium's Fostplus Green Dot program (Fostplus, 2022). If beverage producers pass this cost to consumers, the per tonne cost translates to approximately 1c per PET bottle placed on the market (see calculation in S3.3). This is in addition to a 10c redeemable deposit fee. Previous literature has shown evidence of consumers willing to pay a small price for sustainable packaging (Herrmann et al., 2022) and that deposit fees have not dissuaded consumption so far (Berck et al., 2003).

Low V-PET prices necessitate high EPR fees to fund the increasing cost gap. When the V-PET price is 800 USD/tonne, the fees need to be as high as virgin prices (left-most part of Figure 6). If fees are not sufficiently high, beverage producers and bottle manufacturers may find it cheaper to use virgin material instead of buying R-PET. Policies that ensure high V-PET prices (>1500 USD/tonne) and adherence to recycled content targets (>70%) can help pay for the system without the need for large EPR fees. Eco-modulation of EPR fees based on recycled content is also recommended (Laubinger et al., 2021) to drive demand for bottle-grade R-PET while ensuring the cost of expanded supply is met.

## 4 | DISCUSSION

Many academic studies have reviewed the problems with plastic waste by accounting material flows (Di et al., 2021; Kan et al., 2023), identifying mismanagement routes (Law et al., 2020), and measuring environmental impacts (Borrelle et al., 2020; Cabernard et al., 2022). Researchers, policymakers, and plastic producers advocate for recycling as a partial solution to mitigate these problems (OECD, 2022b; Stegmann et al., 2022; US EPA, 2021). We demonstrate a methodology to quantify the cost of increasing recycling and apply it to PET bottles where technological barriers, relative to other plastics, are minimal. Our results suggest that DRS can improve PET bottle recycling rates to >80% by 2035. Besides diverting waste from disposal, mechanically recycled PET offers an 85% reduction in emissions compared to virgin PET (Uekert et al., 2023). If all the recycled PET is assumed to displace virgin PET use, increased R-PET supply can save 7.6 Mt $CO_2$ eq per year by 2035.

We characterize the US plastics recycling system at greater geographical granularity than previously reported. Much of the context and data that informed our quantitative parametrization of supply and demand variables resides outside peer-reviewed publications. Municipal waste audits, policy response surveys, state-level container refund accounting, trade association publications, corporate sustainability progress reports, and legislative records provide fragmented insights into the current workings of the plastics recycling system at large. Turning science-informed policy discourse into practical solutions requires collating the information scattered in the gray literature. Our bottom-up supply formulation and aggregate demand treatment use this information to estimate net system costs that can inform DRS expansion.

### 4.1 | Limitations

The scope of our study considers the direct economic costs of the recycling system. Indirect but important socioeconomic and environmental considerations such as avoided trash and litter costs (Lavee, 2010; Yu, 2021), the number of jobs created (Bassi et al., 2022), and decreased pollution into water bodies (as much as 40%) (Schuyler et al., 2018), indicate that benefits in practice may be even greater.

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Quality degradation and mismatch are common challenges in circular applications, and our analysis simplifies these considerations for analytical tractability. Studies show there is a significant drop in mechanical and barrier properties of R-PET after several extrusion cycles (Nait-Ali et al., 2011). We do not include quality degradation in once-recycled PET bottles and assume that they can be recycled many times. The literature on PET recycling suggests that more than 50% (up to 90%) recycled content inclusion is dilute enough to not cause severe property degradation after many cycles (Brouwer et al., 2020). However, this article only offers theoretical limits based on empirical observations on contamination, and experimental process validation is lacking. For greater percentage inclusion, chemical recycling or monomer recovery (Chaudhari et al., 2021) which can revert PET back to monomers and yield virgin-like PET, must be considered. With requisite process modeling details, chemical recycling can be included in our analysis. However, due to the lack of consistent data on chemical recycling, it is excluded from the present study.

Two other key assumptions were made to simplify our analysis. First, quality considerations are binary—and reflect the food and non-food grades of R-PET on the market today. However, we also make a strong assumption that curbside collected R-PET bottles are unfit for food-grade purposes due to possible contamination from non-food packaging. While this assumption is based on the 2018 recycling system, the recent spike in demand for R-PET has driven greater investment in sorting single-stream recyclables to get high-quality R-PET (Paben, 2021). Second, we also assume slow growth (on par with population growth) for PET bottles in the United States and do not consider alternative beverage delivery containers (such as aluminum cans) or modes (reusable bottles) that can significantly alter the PET bottle supply and demand accounting.

Circular policies are built on the premise that product circularity (such as bottle-to-bottle recycling) is environmentally superior. This premise derives from studies penalizing downcycling (Huysveld et al., 2022; Rigamonti et al., 2018; Vadenbo et al., 2017) due to quality degradation (Helbig et al., 2022). However, critics emphasize that circularity is a theoretical benchmark (Cullen, 2017) and argue that recycling policies should be guided by the impact reduction potential of recycling activities (Geyer et al., 2016). When processes for upgrading quality (Jehanno et al., 2022) are energy and emissions intensive (Uekert et al., 2023), strategies to maximize value through product circularity may present trade-offs between climate action and waste recovery (Meys et al., 2021). While recycled content mandates can be useful tools to develop R-PET markets, policy incentives for matching supply with demand must consider the broader environmental implications of the system in place.

## 4.2 | Policy recommendations

Our results show that a national DRS with a 10c deposit fee can cost-effectively improve recycling and circularity outcomes. The expanded DRS system incurs a net cost of 360 USD/tonne which translates to approximately 1 cent/bottle on average. Since system costs can increase if manufacturer demand for R-PET is lower than expected, policymakers should combine a national 10c deposit on bottles with demand-side safeguards such as recycled content requirements and eco-modulated EPR fees. DRS expansion and demand-side policies should also account for the loss of recovery value in MRFs so that impacts on the wider waste recovery value chain are appropriately compensated.

Imperfect return rates in DRS programs also generate revenue as unredeemed deposits (i.e., when a customer forfeits deposit value by not returning the PET bottle). When a 10c deposit fee is implemented nationwide, we find that close to 1 billion USD in unredeemed deposits will be potentially available (Figure S11). While this is equivalent to the net system cost for the same scenario, careful policy design is needed to effectively use this revenue without introducing counterproductive or perverse incentives (OECD, 2022a). Current handling of unredeemed deposits varies widely: Connecticut, Massachusetts, and Maine use it for general funds; New York gives producers a fraction to cover system costs while Iowa allows producers to keep it and operate the DRS; Michigan uses it to support retailers who handle deposits, and others like California and Vermont earmark it for specific purposes such as beverage container recycling fund or clean water programs (Reloop Platform, 2020). The degree to which unredeemed funds are allocated back to DRS operations is unclear and must be recognized in future policymaking. A harmonized EPR framework can integrate system costs and revenues and improve accountability. However, several dimensions of EPR implementation, logistics, financing, and material ownership (OECD, 2022a) must be considered to align recycling activity with circular policy goals.

Deposit return systems invite incentivized consumers into the R-PET value chain and expanding such systems nationwide can be a cost-effective PET recycling strategy if robust demand persists. But R-PET demand in the future is uncertain. Rapidly increasing R-PET supply will compete against V-PET and up to 60% of domestic V-PET production—currently estimated to be approximately 4300 kt by Di et al. (2021)—can be potentially displaced. This presents a significant opportunity to decouple from virgin feedstocks in the long term, but low V-PET prices can make recycling expensive during the transition. In this paper, we show that quantifying supply−-demand interaction effects can capture this risk and inform policy instruments to maximize future recovery and circularity potential.

### CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

### DATA AVAILABILITY STATEMENT

Data are available in the article supporting information.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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